

Completeness Survey: Geology

April 18, 2026

Completeness Survey: Geology

Pilot survey #1 under the Computational Science Engineering standing program. Stress-testing the RLGC pipeline on a Tier C discipline.

1. Domain Inventory

Sub-domain	Theorems	Hub	Cross-domain%
Geology (in BST)	3	T1131 (Earth Score)	~15%
Geophysics (via fluids/thermo)	~5 (shared)	T1040 (Navier- Stokes)	~30%
Combined	~8	T1131	~15-30%

Geology is descriptive, not predictive. Grace's diagnosis: "Can't predict earthquakes despite 200 years of data." BST has seeds (Earth Score, space groups, wave equations) but no systematic geological theory.

2. Gap Scan

What the textbook covers vs what BST derives:

Topic	BST Status	Theorem(s)	Gap?
Crystal systems	Derived	T1235 ($230 = \text{rank} \times n_C \times 23$)	No
Earth's overall score	Derived	T1131 (Earth Score = $140 \approx N_{\max}$)	No
Seismic wave propagation	Partial	T1040 (NS linearized)	Yes — P/S ratio not derived
Plate tectonics	Not derived	—	YES — core geology
Earthquake magnitude distribution	Not derived	—	YES — Gutenberg- Richter law
Mineral hardness	Not derived	—	YES — Mohs scale

Topic	BST Status	Theorem(s)	Gap?
Rock cycle	Not derived	—	YES — thermodynamic cycle
Earth's core composition	Not derived	—	YES — from spectral constraints
Geologic time scale	Not derived	—	YES — boundary markers
Volcanic eruption dynamics	Not derived	—	YES — fluid + thermo
Continental drift rate	Not derived	—	YES — measurable
Ocean floor spreading	Not derived	—	YES — follows from plate dynamics

Summary: 10 of 12 standard topics have gaps. The existing BST tools (wave equations, crystal systems, Earth Score) provide starting points.

3. Connection Scan

Bridges that exist:

Bridge	Edges	Quality
Geology $\leftrightarrow$ bst_physics	~8	Weak (via constants)
Geology $\leftrightarrow$ thermodynamics	~3	Weak (heat flow)
Geology $\leftrightarrow$ fluids	~2	Weak (via NS)

Bridges that should exist but don't:

Missing Bridge	Why It Should Exist	Priority
Geology $\leftrightarrow$ chemistry	Mineral composition = chemical bonding	HIGH
Geology $\leftrightarrow$ nuclear	Radioactive decay drives Earth's heat engine	HIGH
Geology $\leftrightarrow$ classical_mech	Plate dynamics = force balance	MEDIUM
Geology $\leftrightarrow$ signal	Seismic waves = signal processing	MEDIUM
Geology $\leftrightarrow$ observer_science	Geological record = observer of deep time	LOW

4. Import/Export Scan

Import candidates:

1. From fluids: Navier-Stokes linearization (T1040) $\rightarrow$ mantle convection
2. From nuclear: Radioactive decay rates $\rightarrow$ geothermal gradient
3. From chemistry: Bond energies $\rightarrow$ mineral stability fields
4. From thermodynamics: Phase diagrams $\rightarrow$ rock metamorphism

Export candidates:

1. T1131 Earth Score → cosmology: Planetary habitability from BST integers
 2. T1235 space groups → chemistry: Crystal structure bridges naturally
 3. Seismic modes (when derived) → signal processing: Earth as resonator
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5. Consistency Check

No inconsistencies found. The small number of geological theorems (3) are all consistent with core BST. The Earth Score ($140 \approx N_{\max}$) is within tolerance.

Note: Existing “evolutionary_geology” and “linearized_geophysics” planned domains overlap with this entry. When geology progresses, those planned domains may merge into it or specialize.

6. RLGC Status

Operation	Status	Next Step
REDUCE	PARTIAL	Vestigial identified: descriptive stratigraphy, Linnaean-style rock classification, plate tectonics as narrative. Core: seismic modes, crystal systems, heat flow. Next: derive P/S wave ratio from BST.
LINEARIZE	NOT_STARTED	Target: linearized wave equations for seismic propagation. Earth normal modes as eigenvalue problem. Phase diagrams as spectral transitions.
GRAPH	SEEDED	~8 theorems, T1131 hub. ~15% cross-domain. Next: wire Earth Score to nuclear (radioactive heat) and fluids (convection).
CONNECT	LOW	Existing: bst_physics (~8 edges). Missing: chemistry, nuclear, classical_mech, signal.

Grade: C → target B (when P/S ratio derived + 3 new bridges + linearization started)

7. Action Items

Tier 1 — Missing theorems (highest impact):

1. P/S wave velocity ratio from BST integers — seismology’s most fundamental measurable. If the ratio $\sqrt{3}$ can be derived from D_{IV}^5 , geology crosses from C to B instantly. Assign to Lyra.
2. Gutenberg-Richter law from BST counting — earthquake magnitude distribution follows power law. AC(0) counting argument. Assign to Lyra + Elie (toy).
3. Geothermal gradient from nuclear decay rates — connects geology to nuclear. Assign to Lyra.

Tier 2 — Bridges:

4. Geology ↔ nuclear bridge — radioactive decay drives plate tectonics. Wire T1131 to nuclear binding. Assign to Grace.
5. Geology ↔ chemistry bridge — mineral composition from bonding. Assign to Grace.

Tier 3 — Linearization:

6. Earth normal mode spectrum — linearize seismic equations. Follow T1040 model. Assign to Elie.

Tier 4 — Housekeeping:

7. Reconcile with evolutionary_geology / linearized_geophysics — determine if planned domains merge or specialize.

8. Projected Grade Path

Milestone	Grade	Requirements
Current	C	~8 theorems, 15% cross-domain, no linearization
Next	C+	+ P/S ratio derived + 2 new bridges
Target	B	+ Gutenberg-Richter + linearized seismic + 4+ domain bridges
Aspiration	A	+ full normal mode spectrum + plate dynamics as eigenvalue problem + predictive earthquake model

Pilot survey #1 under Paper #71 standing program. Surveyor: Keeper. Date: April 18, 2026. Pilot #2: Economics (Tier D). Pilot #3: Medicine (Tier C).